

PC-07 · ACTIVES & VITAMINS · PERSONAL CARE

HS Vitamin C

Ascorbic Acid

INCI	CAS	EC	FAMILY	SUPPLIED AS
Ascorbic Acid	50-81-7	200-066-2	PC-07	Per grade

01 POSITIONING

Antioxidant claims fail on the shelf, not on the skin — we ship the stabilized forms that hold.

02 HOW IT WORKS

- 01

Radical quenching

Tocopherol and ascorbate donate electrons to neutralize the free radicals behind photoaging.
- 02

Brightening pathway

Ascorbic-acid derivatives interfere with melanin synthesis, evening tone over time.
- 03

Stabilized delivery

Ester and phosphate forms (tocopheryl acetate, sodium ascorbyl phosphate) resist oxidation until skin enzymes release the active.

03 APPLICATIONS

- Antioxidant serums and creams
- Brightening and even-tone lines
- After-sun and pollution-defense
- Hair and scalp actives
- Nutraceutical premixes (grade-dependent)

04 FORMULATION GUIDE

USE LEVEL	0.2–2%
PH & CONDITIONS	Form-dependent: ascorbyl phosphate pH 6–7; tocopheryl acetate pH-flexible; free acids pH < 3.5
— Add antioxidants in the cool-down phase; protect ascorbates from metal ions with a chelant.	
— Prefer stabilized ester/phosphate forms for aqueous shelf-stable products.	

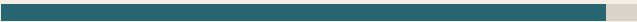
05 TYPICAL SPECIFICATION

Appearance	per grade: oil / powder / crystals
Assay	≥ 95–98%
Grade	cosmetic / USP-adjacent per TDS

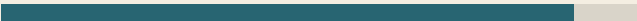
Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Shelf stability of stabilized vs free form
Ascorbyl phosphate / tocopheryl acetate



Antioxidant capacity (class data)
Vitamin systems



Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.